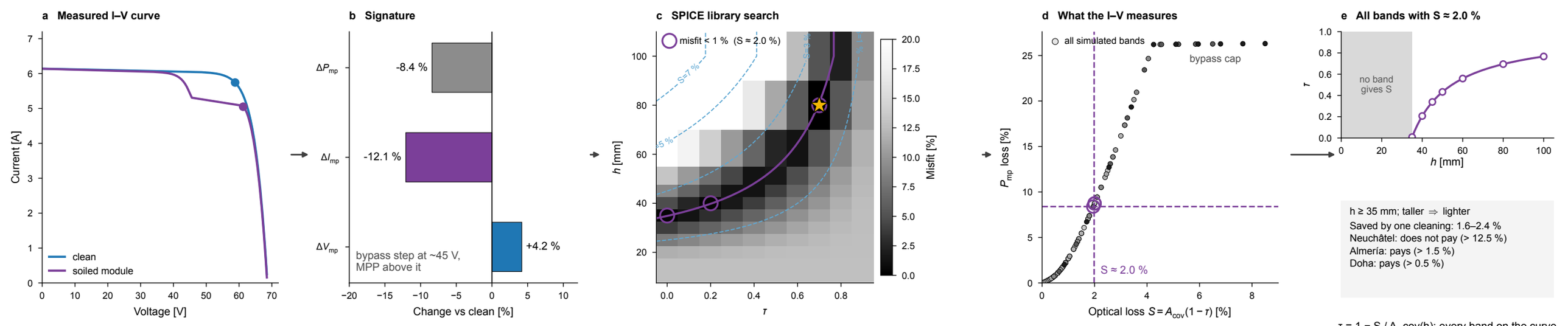

a Measured I–V curve
Current [A]
Voltage [V]
clean
soiled module
b Signature
ΔP_mp
ΔI_mp
ΔV_mp
-8.4 %
-12.1 %
+4.2 %
bypass step at ~45 V,
MPP above it
Change vs clean [%]
c SPICE library search
misfit < 1 % (S ≈ 2.0 %)
S=7 %
S=5 %
S=3 %
S=1 %
h [mm]
τ
Misfit [%]
d What the I–V measures
all simulated bands
bypass cap
P_mp loss [%]
S ≈ 2.0 %
Optical loss S = A_cov(1 − τ) [%]
e All bands with S ≈ 2.0 %
no band
gives S
τ
h [mm]
h ≥ 35 mm; taller ⇒ lighter
Saved by one cleaning: 1.6–2.4 %
Neuchâtel: does not pay (> 12.5 %)
Almería: pays (> 1.5 %)
Doha: pays (> 0.5 %)
IV tracer on a soiled module
(SunPower, landscape) vs clean
reference; the band is unknown
relative change of P_mp, I_mp, V_mp
and bypass step (yes / no)
misfit to every simulated band;
⋆ closest, ○ within 1 point:
they lie on one iso-S line
all bands collapse on one curve:
the I–V gives S,
not h and τ separately
τ = 1 − S / A_cov(h): every band on the curve
fits the I–V; same decision for all
IV tracer (here: SPICE curve)
Python
SPICE library (LTspice, precomputed)
Python + cleaning map